# HW4 for CS 6210

You may (and probably should) talk about problems with the each other, with the TA, and with me, providing attribution for any good ideas you might get. Your final write-up should be your own.

```
using LinearAlgebra
```

```
using Plots
```

```
using FastGaussQuadrature
```

# 1. Tree climbing

Consider a positive definite matrix whose graph structure is a tree, represented as three arrays:

- An array of ancestor indices `p`, i.e. `p[i]` is the parent of node `i`. Assume ordering such that `p[i] > i`, and use the convention that the last node is the root (and the parent doesn't matter).
- An array of diagonal values `d`
- An array of off-diagonal values `w` (where `w[i]` is the weight between `i` and `p[i]`)

Fill in a code to compute new arrays `dr` and `wr` associated with the Cholesky factor.

```
tree_chol (generic function with 1 method)
```

```
function tree_chol(p, d, w)
    n = length(p)
    dr = copy(d)
    wr = copy(w)
    for i = 1:n-1
        dr[i] = sqrt(dr[i])
        wr[i] = wr[i] / dr[i]
        dr[p[i]] -= wr[i]^2
    end
    dr[n] = sqrt(dr[n])

    dr, wr
end
```

```
([2.23607, 2.44949, 2.63787, 2.82843, 3.09436], [0.0, -0.204124, -0.379094, -0.53033])
```

```
let
    p = [3; 3; 5; 5; -1]
    d = [5.0; 6.0; 7.0; 8.0; 10.0]
    w = [0.0; -0.5; -1.0; -1.5]
    tree_chol(p, d, w)
end
```

# 2. Somewhat sensitive

Using the standard formulation

$$\begin{bmatrix} I & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} r \\ x \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix},$$

compute the variation of the solution $\delta x$ with respect to variations $\delta A$ of the matrix. Assume that $A \in \mathbb{R}^{m \times n}$ with $m > n$, and that $A$ has full column rank. Implement as the code

```
function deriv_ls(A, b, δA)
    # Returns the variation of δx w.r.t. variation δA
end
```

Then, $r = b - Ax$. Perturbed: $\begin{bmatrix} I & A+\delta A \\ (A+\delta A)^T & 0 \end{bmatrix} \begin{bmatrix} r+\delta r \\ x+\delta x \end{bmatrix} \approx \begin{bmatrix} b \\ 0 \end{bmatrix}$

$$\delta r + A\delta x = -\delta A x$$

$$A^T \delta r = -\delta A^T r$$

$$A^T A \delta x = \delta A^T r - A^T(\delta A x)$$

$$\delta x = (A^T A)^{-1}(\delta A^T r - A^T \delta A x)$$

condition number $\kappa(A)$ bounds $\frac{||\delta x||}{||x||} \leq \kappa(A) \frac{||\delta A||}{||A||} + O(||\delta A||^2)$

You may use the following sanity check code to test your computation.

```
deriv_ls (generic function with 1 method)
```

```
function deriv_ls(A, b, δA)
    x = A \ b
    r = b - A * x
    δx = (A' * A) \ (δA' * r - A' * (δA * x))

    δx
end
```

```
2.293888965913124e-10
```

```
let
    A = rand(10,3)
    δA = rand(10,3)
    b = rand(10)
    h = 1e-6
    xp = (A+h*δA)\b
    xm = (A-h*δA)\b
    δx_fd = (xp-xm)/2/h
    δx_an = deriv_ls(A, b, δA)
    norm(δx_fd-δx_an)/norm(δx_fd)
end
```

# 3. Pesky probability

Suppose $Z$ is a standard normal random variable. Compute the coefficients for the degree eight polynomial

$$p(x) = \sum_{j=1}^{5} c_j x^{2(j-1)}$$

that minimizes $E[(p(Z) - \cos(Z))^2]$. Use the Gauss-Hermite rule from `FastGaussQuadrature` to compute any relevant integrals (an example is given below).

This is $L^2$ projection onto $\{1, x^2, x^4, x^6, x^8\}$ under Gaussian measure. The goal is to find coefficients $c_1, c_2, c_3, c_4, c_5$ for the even polynomial $p(x) = c_1 + c_2 x^2 + c_3 x^4 + c_4 x^6 + c_5 x^8$ that minimizes $\mathbb{E}[(p(Z) - \cos(Z))^2]$, where $Z \sim \mathcal{N}(0, 1)$.

This is equivalent to projecting $\cos(Z)$ onto the subspace spanned by $\{1, Z^2, Z^4, Z^6, Z^8\}$ under the inner product $\langle f, g \rangle = E[f(Z)g(Z)]$. The solution satisfies the normal equations $G\mathbf{c} = h$, where:$G_{jk} = E[Z^{2(j-1)+2(k-1)}]$ (Gram matrix of even moments), $h_j = E[\cos(Z) Z^{2(j-1)}]$.

These expectations are computed using Gauss-Hermite quadrature, which approximates integrals against $e^{-x^2}$. To match the normal density $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, scale the nodes and weights: nodes $x_i = \sqrt{2} Z_i$, weights $w_i / \sqrt{\pi}$, where $(Z_i, w_i)$ are from gausshermite(50)

From the computation: $c = [0.999828, -0.499124, 0.0410672, -0.00126361, 1.50429e-5]$.

```
pesky_prob_coeff (generic function with 1 method)
```

```
function pesky_prob_coeff()
    n = 5
    Z, w = gausshermite(50)
    x = sqrt(2) .* Z
    w_adjusted = w / sqrt(pi)
    even_comp = 2 * (0:4)
    G = zeros(n, n)
    for j = 1:n
        for k = 1:n
            exp_jk = even_comp[j] + even_comp[k]
            G[j, k] = dot(w_adjusted, x .^ exp_jk)
        end
    end

    h = zeros(n)
    for j = 1:n
        exp_j = even_comp[j]
        h[j] = dot(w_adjusted, cos.(x) .* (x .^ exp_j))
    end

    c = G \ h
    c
end
```

```
coeff =  [0.999828, -0.499124, 0.0410672, -0.00126361, 1.50429e-5]
```

```
coeff = pesky_prob_coeff()
```

```
1.3803884470431427
```

```
let
    # Ex: Compute integral cos(x) ϕ(x) dx for ϕ(x) the standard normal PDF
    Z, w = gausshermite(50)
    dot(w, cos.(Z))
end
```